# A Degenerated Continuum Rayleigh-Ritz Framework for the Free Vibration and Stability of Rotating Compressor Blade

Or

Free Vibration Analysis of a Rotating, Pre-twisted, Cambered, and Tapered NACA 6412 Airfoil Compressor Blade

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## Abstract

A semi-analytical framework is developed for the free vibration and dynamic stability analysis of rotating compressor blades modelled as pre-twisted, tapered, and cambered NACA four-digit airfoil shells. The blade geometry is described using an exact parametric mid-surface mapping with analytically evaluated derivatives, while three-dimensional strains are obtained from the displacement gradient using the inverse mapping Jacobian. A five-field first-order shear deformation model is discretized using the Rayleigh–Ritz method with Jacobi and Legendre polynomial basis functions. Singularities associated with the NACA leading-edge thickness and discontinuities in camber curvature are addressed through specialized coordinate transformations and domain subdivision.

Rotational effects are incorporated through centrifugal prestress, geometric stiffness, Coriolis coupling, and spin-softening. The resulting quadratic eigenvalue problem is solved using an energy-scaled formulation to improve numerical stability and capture divergence-type instabilities. The framework is validated against three-dimensional finite element analyses (ANSYS) and published benchmarks.

The integrated blade mass and volume agree with three-dimensional CAD models within 0.22%. A (6×6) Ritz order (180 degrees of freedom) predicts natural frequencies within 0.07% for the fundamental mode and 0.88% across the primary modes at rotational speeds up to 4,000 rpm. At speeds up to 20,000 rpm, the maximum frequency error remains below 0.84%. Comparisons with published benchmarks yield an RMSE below 0.43%, compared with 1.88% for existing formulations. The proposed framework provides an accurate and computationally efficient approach to predicting the vibration characteristics and dynamic stability of rotating compressor blades.

**Keywords:** *rotating blade; pre-twist; degenerated shell; first-order shear deformation theory; Rayleigh–Ritz; Jacobi polynomials; centrifugal stiffening; Coriolis effect; gyroscopic eigenvalue problem; dynamic stability; NACA airfoil.*

## 1. Introduction

### 1.1. Background and Motivation

Compressor and turbine blades operate in a rotating frame under centrifugal loads that alter their stiffness, couple their bending and torsional motions, and, through Coriolis acceleration, render the governing operator gyroscopic. Accurate prediction of the rotating natural frequencies and of the Campbell diagram is a prerequisite for avoiding resonant engine orders and for high-cycle-fatigue assessment [1–4]. The subject has a long history. Southwell and Gough [5] gave the first estimate of centrifugal stiffening for airscrew blades. Carnegie [6,7] formulated energy expressions for pre-twisted rotating cantilever blading, and Rao [8] consolidated the beam-based literature. Beam models with pre-twist, taper, and Coriolis coupling have since been refined extensively [9–17], and surveys are available [18,19].

Beam theory is inadequate when the blade aspect ratio is low, the chordwise flexibility is significant, or camber and thickness variation across the chord control the modal content. Plate and shell descriptions were therefore introduced. Dokainish and Rawtani [20] analyzed rotating cantilever plates by finite elements; Sreenivasamurthy and Ramamurti [21] examined the Coriolis effect on rotating low-aspect-ratio plates; Ramamurti and Kielb [22] treated twisted

rotating plates; Leissa and co-workers [23–26] developed Ritz solutions for cantilevered shallow shells, twisted rotating blades, and three-dimensional twisted parallelepipeds; Yoo and co-workers [27,28] derived modal characteristics of rotating rectangular plates; Hu et al. [29], Kee and Kim [30], Sinha and Turner [31], and Sun et al. [32,33] extended shell models to pre-twisted, cambered, and staggered blades. Reviews of shell vibration [34,35] document the breadth of this literature.

### 1.2. Literature Review and Identified Research Gap

Two limitations recur in the shell-based rotating-blade literature. First, most of the analytical and semi-analytical shell formulations rest on classical two-dimensional shell theories (Love–Kirchhoff, Donnell, Sanders, Reissner–Mindlin resultant theories) [36–39]. These theories express the metric through Lamé parameters $A_1, A_2$ and principal curvatures $R_1, R_2$, which are available in closed form only for canonical geometries (cylindrical, conical, spherical, shallow doubly curved panels). A pre-twisted, tapered, cambered airfoil surface has non-orthogonal parametric lines, non-constant Gaussian curvature, and a thickness that varies in both directions. Fitting such a surface into a Lamé-parameter framework requires either approximating the geometry (shallow-shell or "twisted plate" assumptions) or deriving lengthy problem-specific curvature expressions that are prone to error and do not generalize.

Second, formulations that do treat the geometry exactly, the degenerated-continuum shell elements of Ahmad, Irons and Zienkiewicz [40] and their descendants [41-45] are implemented almost exclusively within the finite element method. The geometric exactness is achieved by interpolating nodal coordinates and nodal directors, which ties the formulation to a mesh, introduces the well-known locking pathologies of low-order shell elements, and forfeits the compactness and convergence rate of global approximation.

The Rayleigh–Ritz method with orthogonal polynomial admissible functions [46–50] delivers exponential convergence for smooth problems with a small number of degrees of freedom and has been applied successfully to plates and shells of canonical geometry. Its use for rotating blades has, however, remained coupled to classical shell kinematics, and its potential in combination with an exact three-dimensional mapping has not been exploited.

### 1.3. Proposed Approach and Key Contributions

This paper closes the gap by combining the two ingredients. The blade is described as a degenerated three-dimensional continuum: the mid-surface position vector and its analytic derivatives, the exact unit normal and its derivatives, and the through-thickness offset define a three-dimensional Jacobian $\mathbf{J}(x, \eta, \zeta)$ that is inverted at every quadrature point. Strains are then the components of the exact Cartesian displacement gradient, rotated into a local orthonormal frame, with the transverse normal strain removed by the plane-stress condition. The displacement field follows a five-parameter first-order shear deformation theory (FSDT) [36,37,51]: three global Cartesian translations of the mid-surface and two director rotations expressed in the local frame. The unknown fields are expanded globally over the whole blade with $L_2$-orthonormal Jacobi–Legendre tensor products; the span functions vanish identically at the root, so the clamped condition is satisfied exactly for all five fields.

Rotation is treated in the blade-fixed frame [52–54]. The centrifugal body force is integrated consistently to a load vector, a three-dimensional static equilibrium problem is solved, the resulting Cauchy stress tensor is evaluated at every volume quadrature point, and the initial-stress (geometric) stiffness matrix is assembled from it. The Coriolis acceleration produces a skew-symmetric gyroscopic matrix and the centripetal acceleration a spin-softening matrix. The quadratic eigenvalue problem is solved by an energy-scaled reformulation that exploits the structure of conservative gyroscopic systems [55–58].

The principal contributions of this work are summarized as follows:

i. We derived a Lamé-parameter-free degenerated continuum shell formulation for pre-twisted, tapered, and cambered NACA airfoil compressor blades, wherein the complete 3D differential geometry enters strictly through the continuous mid-surface position vector $\mathbf{r}(x, \eta)$ and its analytical derivatives, bypassing the geometric approximations and coordinate limitations of classical shell theories.
ii. We formulated a single-patch Rayleigh–Ritz discretization using $L_2$-orthonormal Jacobi and Legendre polynomials, combining exact clamped-root boundary enforcement with a singularity-aware 3D volume quadrature scheme ($\eta = t^2$ coordinate transformation and maximum-camber domain split) that restores spectral convergence across singular leading-edge thickness gradients and camber slope discontinuities.
iii. We established a prestress-consistent treatment of centrifugal stiffening, in which the full three-dimensional Cauchy stress tensor $\mathbf{\Sigma}_0$ recovered from a static equilibrium solve directly drives the initial-stress geometric stiffness matrix $\mathbf{K}_\sigma$ using the identical kinematic displacement operators and Jacobian determinants.

iv. We implemented an energy-scaled Hermitian gyroscopic eigensolver that exploits the underlying state-space Hamiltonian structure to guarantee purely imaginary eigenvalues ($\mathrm{Re}(\lambda) = 0$) in the stable regime, while seamlessly transitioning to a time-scaled companion solver for divergence boundary detection ($\mathrm{Re}(\lambda) > 0$) with residual-based verification ($r \leq 10^{-6}$) for every computed eigenpair.

The remainder of this paper is structured as follows. Section 2 establishes the analytical differential geometric mapping for the tapered, pre-twisted, and curved NACA 6412 blade mid-surface. Section 3 details the FSDT shell kinematics, the Rayleigh–Ritz basis functions, the rotating energy functionals, the 3D singularity-aware quadrature scheme, and the energy-scaled Hermitian eigensolver. Section 4 presents the computational implementation, vectorized matrix assembly, software stack, and baseline reference blade configuration. Section 5 evaluates the framework through geometric verification, numerical convergence testing, non-rotating and rotating finite element validations against 3D ANSYS models, and parametric Campbell diagram stability analyses. Finally, Section 6 summarizes the primary findings and conclusions.

## 2. Blade Geometry and Differential Mapping

### 2.1. Reference Coordinate Systems and Frames

A right-handed global Cartesian frame $(X, Y, Z)$ with unit vectors $\mathbf{e}_X, \mathbf{e}_Y, \mathbf{e}_Z$ is fixed to the rotor and rotates with it at constant angular velocity $\boldsymbol{\Omega} = \Omega\,\mathbf{e}_Y$, $\Omega = 2\pi N/60$, where N is the rotor speed in rpm. The blade span is directed along X (radial direction), starting at the hub radius $R_h$. The stacking (pre-twist) axis is the radial line $Y = Z = 0$. The mid-surface is parametrized by the span coordinate $x \in [0, L]$ and the dimensionless chord coordinate $\eta \in [0,1]$, measured from the leading edge ($\eta = 0$) to the trailing edge ($\eta = 1$). The through-thickness coordinate is $\zeta \in [-h/2\,,\, h/2]$.

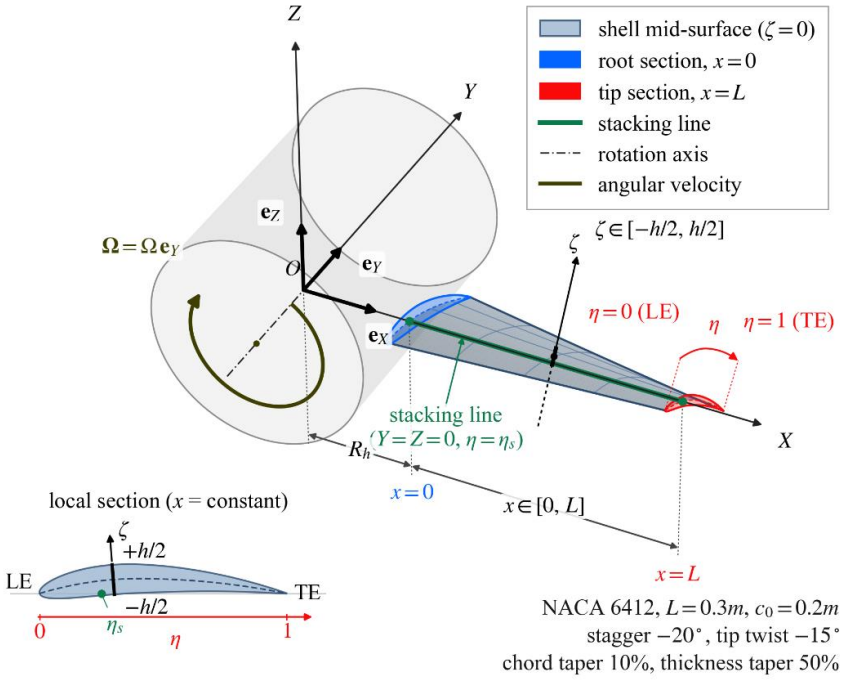



Figure 1. Rotating coordinate systems and mid-surface geometric mapping: Global rotating reference frame $(X, Y, Z)$ with angular velocity $\boldsymbol{\Omega} = \Omega\mathbf{e}_Y$, radial hub radius $R_h$, and local shell coordinates $(x, \eta, \zeta)$ defined on the pre-twisted compressor blade mid-surface.

### 2.2. NACA Airfoil Geometry and Camber Discontinuity

The section profile is a NACA four-digit airfoil [59, 60] specified by maximum camber m (first digit, percent of chord), camber position p (second digit, tenths of chord), and maximum thickness-to-chord ratio $\tau$ (last two digits, percent of chord). The camber line $f(\eta)$ and its second derivative $f''(\eta)$ are given by:

$$f(\eta) = \begin{cases} \dfrac{m}{p^2}(2p\eta - \eta^2), & \eta \le p \\ \dfrac{m}{(1-p)^2}(1 - 2p + 2p\eta - \eta^2), & \eta > p \end{cases}, \qquad f''(\eta) = \begin{cases} -\dfrac{2m}{p^2}, & \eta \le p \\ -\dfrac{2m}{(1-p)^2}, & \eta > p \end{cases}, \tag{1}$$

so that $f(\eta)$and $f'(\eta)$are continuous across the chord, whereas $f''(\eta)$ exhibits a finite jump discontinuity at $\eta = p$. The section thickness distribution $h_0(\eta)$ is defined as:

$$h_0(\eta) = 10\,\tau\,c\big(0.2969\sqrt{\eta} - 0.1260\,\eta - 0.3516\,\eta^2 + 0.2843\,\eta^3 + a_4\,\eta^4\big), \tag{2}$$

with $a_4 = -0.1015$ for the standard finite trailing-edge thickness and $a_4 = -0.1036$ for a closed trailing edge. The $\sqrt{\eta}$ term produces the rounded leading edge and introduces an integrable singularity in $\partial h_0/\partial\eta$ at $\eta = 0$.

### 2.3. Spanwise Taper, Pre-twist, and Stacking Axis

The section chord tapers linearly along the span according to $c(x) = c_0(1 - \tau_c x/L)$, while the thickness is scaled independently as $h(x,\eta) = h_0(x,\eta)(1 - \tau_h x/L)$, where $c_0$ is the root chord length, and $\tau_c, \tau_h \in [0, 1)$ denote the non-dimensional chord and thickness taper fractions, respectively.

Each cross-section is rotated about the spanwise stacking axis by the linearly varying angle:

$$\theta(x) = \theta_0 + \kappa x, \qquad \kappa = \frac{\theta_{tip}}{L}, \tag{3}$$

where $\theta_0$ is the root stagger angle, $\theta_{tip}$ is the total relative tip pre-twist, and $\kappa$ represents the constant spanwise twist rate. The corresponding section-fixed unit vectors $\mathbf{e}_y(x)$ and $\mathbf{e}_z(x)$ in the rotating frame are given by:

$$\mathbf{e_y}(x) = \begin{bmatrix} 0 \\ \cos\theta\,(x) \\ \sin\theta\,(x) \end{bmatrix}, \qquad \mathbf{e_z}(x) = \begin{bmatrix} 0 \\ -\sin\theta\,(x) \\ \cos\theta\,(x) \end{bmatrix}. \tag{4}$$

The radial stacking axis pierces each section on its camber line at the non-dimensional chord fraction $\eta_a$. The camber line ordinate is shifted so that this point coincides with the origin of the local section plane, $\hat{f}(\eta) = f(\eta) - f(\eta_a)$, and the chordwise offset from the stacking axis is defined as $a(\eta) = \eta - \eta_a$.

### 2.4. Analytical Mid-surface Parametrically Mapped Derivatives

The 3D mid-surface position vector $\mathbf{r}(x,\eta)$ of any point on the blade shell is given by:

$$\mathbf{r}(x,\eta) = (R_h + x)\,\mathbf{e}_X + c(x)\big[a(\eta)\,\mathbf{e}_y(x) + \hat{f}(\eta)\,\mathbf{e}_z(x)\big]. \tag{5}$$

To streamline partial differentiation, two orthogonal section auxiliary vectors $\mathbf{v}(x,\eta)$ and $\mathbf{w}(x,\eta)$ are introduced:

$$\mathbf{v}(x,\eta) = a(\eta)\mathbf{e_y}(x) + \hat{f}(\eta)\mathbf{e_z}(x), \qquad \mathbf{w}(x,\eta) = a(\eta)\mathbf{e_z}(x) - \hat{f}(\eta)\mathbf{e_y}(x). \tag{6}$$

Because the section triad differentiates as $\partial\mathbf{e_y}/\partial x = \kappa\mathbf{e_z}$ and $\partial\mathbf{e_z}/\partial x = -\kappa\mathbf{e_y}$, these auxiliary vectors satisfy $\partial\mathbf{v}/\partial x = \kappa\mathbf{w}$ and $\partial\mathbf{w}/\partial x = -\kappa\mathbf{v}$.

The first covariant mid-surface tangent vectors are evaluated analytically as:

$$\mathbf{r}_{,x} = \frac{\partial\mathbf{r}}{\partial x} = \mathbf{e_X} + c'(x)\mathbf{v} + c(x)\kappa\mathbf{w}, \qquad \mathbf{r}_{,\eta} = \frac{\partial\mathbf{r}}{\partial\eta} = c(x)\big(\mathbf{e_y} + f'(\eta)\mathbf{e_z}\big), \tag{7}$$

where $c'(x) = -c_0\tau_c/L$ is the constant chord taper derivative along the span. Differentiating Eq. (7) yields the second partial surface derivatives:

$$\begin{gathered} \mathbf{r}_{,xx} = 2c'(x)\kappa\mathbf{w} - c(x)\kappa^2\mathbf{v} \\ \mathbf{r}_{,x\eta} = c'(x)\big(\mathbf{e_y} + f'(\eta)\mathbf{e_z}\big) + c(x)\kappa\big(\mathbf{e_z} - f'(\eta)\mathbf{e_y}\big) \\ \mathbf{r}_{,\eta\eta} = c(x)f''(\eta)\mathbf{e_z}. \end{gathered} \tag{8}$$

The unnormalized surface normal vector is $\mathbf{g} = \mathbf{r}_{,x} \times \mathbf{r}_{,\eta}$, the surface Jacobian determinant is $j = \| \mathbf{g} \| > 0$, and the unit normal vector $\mathbf{n}$ is:

$$\mathbf{n} = \frac{\mathbf{r}_{,x} \times \mathbf{r}_{,\eta}}{j}. \tag{9}$$

Differentiating Eq. (9) with respect to surface parametric coordinates $\alpha \in \{x, \eta\}$ and projecting out the normal component yields the spatial derivatives of the unit normal vector:

$$\mathbf{n}_{,\alpha} = \frac{\mathbf{1}}{j}\left[\mathbf{g}_{,\alpha} - \mathbf{n}\left(\mathbf{n} \cdot \mathbf{g}_{,\alpha}\right)\right] \tag{10}$$

where the partial derivatives of the unnormalized surface normal cross-product $\mathbf{g}_{,x}$ and $\mathbf{g}_{,\eta}$ are given by:

$$\mathbf{g}_{,x} = \mathbf{r}_{,xx} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,x\eta}, \qquad \mathbf{g}_{,\eta} = \mathbf{r}_{,x\eta} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,\eta\eta}. \tag{11}$$

### 2.5. Local Orthonormal Surface Frame and Analytic Frame Derivatives

At every mid-surface point, a right-handed orthonormal triad $\mathbf{Q} = [\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3]$ is constructed with

$$\mathbf{e}_1 = \frac{\mathbf{r}_{,x}}{\| \mathbf{r}_{,x} \|}, \qquad \mathbf{e}_3 = \mathbf{n}, \qquad \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_1. \tag{12}$$

The spatial partial derivatives of the frame $\mathbf{Q}_{,x}$ and $\mathbf{Q}_{,\eta}$(or $\mathbf{Q}_{,\alpha}$ for $\alpha \in \{x, \eta\}$) are evaluated analytically using the same projection operator applied in Eq. (10) alongside the product rule for the cross product:

$$\mathbf{e}_{1,\alpha} = \frac{1}{\| \mathbf{r}_{,x} \|}(\mathbf{I} - \mathbf{e}_1\mathbf{e}_1^{\mathrm{T}})\mathbf{r}_{,x\alpha}, \qquad \mathbf{e}_{3,\alpha} = \mathbf{n}_{,\alpha}, \qquad \mathbf{e}_{2,\alpha} = \mathbf{n}_{,\alpha} \times \mathbf{e}_1 + \mathbf{n} \times \mathbf{e}_{1,\alpha}, \tag{13}$$

where $\mathbf{I}$ is the $3 \times 3$ identity matrix and $\mathbf{e}_1\mathbf{e}_1^{\mathrm{T}}$ denotes the outer product projection tensor.

The availability of $\mathbf{Q}_{,\alpha}$ in exact closed form is essential to the degenerated continuum formulation: the director rotations of the First-Order Shear Deformation Theory (FSDT) kinematics are resolved directly within this local frame, and their spatial derivatives require the spatial derivatives of the frame itself. Consequently, no Lamé parameters $(A_1, A_2)$, Christoffel symbols, or principal curvature tensors $(R_1, R_2)$ appear anywhere in the formulation. The shell geometry is specified entirely by the mid-surface position vector $\mathbf{r}$ and its first and second analytic derivatives $\left(\mathbf{r}_{,\alpha}, \mathbf{r}_{,\alpha\beta}\right)$.

## 3. Mathematical Formulation

### 3.1. Degenerated Continuum Kinematics and FSDT Displacement Field

3.1.1. Degenerated 3D Continuum Geometry and Velocity Field

The 3D spatial position vector $\mathbf{X}(x, \eta, \zeta)$ of any material point within the solid blade volume in the reference configuration is expressed as:

$$\mathbf{X}(x, \eta, \zeta) = \mathbf{r}(x, \eta) + \zeta\, \mathbf{n}(x, \eta), \tag{14}$$

where $\mathbf{r}(x, \eta)$ is the mid-surface position vector, $\mathbf{n}(x, \eta)$ is the unit surface normal, and $\zeta \in [-h/2, h/2]$ is the through-thickness coordinate.

The exact 3D Jacobian matrix $\mathbf{J}$ of the parametric mapping $(x, \eta, \zeta) \mapsto \mathbf{X}$ is evaluated by differentiating Eq. (14) with respect to $(x, \eta, \zeta)$:

$$\mathbf{J} = \left[\, \mathbf{r}_{,x} + \zeta\, \mathbf{n}_{,x} \middle|\, \mathbf{r}_{,\eta} + \zeta\, \mathbf{n}_{,\eta} \middle|\, \mathbf{n} \,\right], \qquad \det \mathbf{J} > 0. \tag{15}$$

The determinant $\det \mathbf{J}$ represents the local volume transformation ratio from the parametric domain to 3D physical space. In the numerical solver, $\det \mathbf{J}$ is monitored at every 3D quadrature point; a non-positive value $(\det \mathbf{J} \leq 0)$indicates that the local shell thickness exceeds the minimum radius of curvature, signaling geometric self-intersection where the degenerated continuum description loses physical validity.

3.1.2. Five-Field FSDT Displacement Field and Spatial Strain Operators

The displacement field follows a five-parameter First-Order Shear Deformation Theory (FSDT) [36,37,51]:

$$\mathbf{u}(x, \eta, \zeta) = \mathbf{u}_0(x, \eta) + \zeta[\phi_1(x, \eta)\, \mathbf{e}_1(x, \eta) + \phi_2(x, \eta)\, \mathbf{e}_2(x, \eta)], \tag{16}$$

where $\mathbf{u}_0 = (u, v, w)^T$ denotes the vector of global Cartesian translations of the mid-surface, and $\phi_1, \phi_2$ are the director rotations resolved in the local tangent frame $(\mathbf{e_1}, \mathbf{e_2})$. The drilling rotation is omitted, and the director is inextensible ($\varepsilon_{33} = 0$), which is consistent with the plane-stress material constitutive law adopted below.

The partial derivatives of the displacement field with respect to the parametric coordinates are evaluated analytically as:

$$\mathbf{u}_{,x} = \mathbf{u}_{0,x} + \zeta\left(\phi_{1,x}\mathbf{e}_1 + \phi_1\mathbf{e}_{1,x} + \phi_{2,x}\mathbf{e}_2 + \phi_2\mathbf{e}_{2,x}\right), \qquad \mathbf{u}_{,\zeta} = \phi_1\mathbf{e}_1 + \phi_2\mathbf{e}_2, \tag{17}$$

with the spanwise derivative $\mathbf{u}_{,\eta}$ defined analogously. The frame-derivative terms $\phi_\alpha \mathbf{e}_{\alpha,x}$ and $\phi_\alpha \mathbf{e}_{\alpha,\eta}$ capture the exact kinematic coupling between director rotations and the 3D twist, taper, and camber of the shell surface. These are precisely the terms that classical shell theories approximate through assumed curvature tensors and Lamé parameters.

The global 3D Cartesian displacement gradient tensor $\nabla_X \mathbf{u}$ is evaluated using the inverse 3D Jacobian:

$$\nabla_X \mathbf{u} = \left[\mathbf{u}_{,x}\ \mathbf{u}_{,\eta}\ \mathbf{u}_{,\zeta}\right]\mathbf{J}^{-1}. \tag{18}$$

Transforming this gradient into the local orthonormal material frame $\mathbf{Q} = [\mathbf{e}_1\ \ \mathbf{e}_2\ \ \mathbf{e}_3]$ yields the local displacement gradient matrix $\mathbf{H}$:

$$\mathbf{H} = \mathbf{Q}^T(\nabla_X \mathbf{u})\mathbf{Q}. \tag{19}$$

In the computational implementation, the product $\mathbf{J}^{-1}\mathbf{Q}$ is evaluated via one $3 \times 3$ linear system solve per quadrature point, after which the spatial gradient along the local material directions is directly available for constructing both the strain–displacement operator matrix and the initial-stress geometric stiffness matrix.

The 5-component 3D engineering strain vector $\boldsymbol{\varepsilon}$ used with the plane-stress constitutive law is extracted from $\mathbf{H}$ as:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \\ \gamma_{13} \\ \gamma_{23} \end{bmatrix} = \begin{bmatrix} H_{11} \\ H_{22} \\ H_{12} + H_{21} \\ H_{13} + H_{31} \\ H_{23} + H_{32} \end{bmatrix} = \mathbf{B}(x, \eta, \zeta)\mathbf{q}\,, \tag{20}$$

where $\mathbf{q}$ is the global vector of Ritz expansion coefficients, and $\mathbf{B}(x, \eta, \zeta)$ is the spatial strain–displacement operator matrix. The transverse normal strain $\varepsilon_{33}$ is eliminated by enforcing the plane-stress condition $\sigma_{33} = 0$.

Because $\zeta$ is treated as an independent coordinate in the spatial differentiation of Eq. (16), the spanwise and chordwise variation of thickness enters exclusively through the integration domain limits and the volume determinant $\det \mathbf{J}$, rather than through artificial derivatives in the strain operator, a distinction that is automatic in the degenerated continuum formulation but must be imposed manually in resultant-based shell theories.

### 3.2. Orthonormal Jacobi–Legendre Rayleigh–Ritz Basis Functions

Each of the five generalized displacement fields $\{u, v, w, \varphi_1, \varphi_2\}$ is expanded in the same tensor-product basis of $n_s \times n_c$ functions:

$$\psi_{ij}(x, \eta) = X_i(x)\, Y_j(\eta), \qquad i = 0, \ldots, n_s - 1, \quad j = 0, \ldots, n_c - 1, \tag{21}$$

yielding a total of $n = 5n_s n_c$ global degrees of freedom.

#### 3.2.1. Spanwise Basis Functions and Exact Clamped Enforcement

Mapping the physical spanwise coordinate $x \in [0, L]$ to the non-dimensional interval $s = 2x/L - 1 \in [-1, 1]$, the spanwise basis functions are defined as:

$$X_i(x) = \sqrt{\frac{2i + 3}{8}}\ (1 + s)\, P_i^{(0,2)}(s), \tag{22}$$

where $P_i^{(\alpha,\beta)}(s)$ is the Jacobi polynomial [65,66]. The boundary factor $(1 + s)$ vanishes identically at the root $x = 0$ (where $s = -1$), ensuring that every field (translations and director rotations alike) satisfies the fully clamped kinematic boundary condition identically, without requiring penalty functions or artificial spring constraint equations.

The Jacobi weight function $(1+s)^{\beta}$ with $\beta = 2$ naturally absorbs the square of this boundary factor, yielding exact orthonormality:

$$\int_{-1}^{1} X_i(s)\, X_k(s)\, ds = \delta_{ik}, \tag{23}$$

by virtue of the Jacobi polynomial orthogonality relation:

$$\int_{-1}^{1} (1+s)^2 P_i^{(0,2)}(s) P_k^{(0,2)}(s)\, ds = \frac{8}{2i+3}\delta_{ik}.$$

The spatial derivative with respect to s is evaluated analytically using:

$$\frac{d}{ds} P_i^{(0,2)}(s) = \frac{1}{2}(i+3) P_{i-1}^{(1,3)}(s).$$

The spanwise basis thus forms a set of $L_2$-orthonormal admissible functions in the sense of Bhat [46] and Liew and Lam [47]. Because these functions are obtained in exact closed form rather than via Gram–Schmidt numerical orthogonalization, they avoid the loss of orthogonality and numerical ill-conditioning that accumulate in generated recurrence sequences [50].

3.2.2. Chordwise Basis Functions and Free Boundary Conditions

Mapping the non-dimensional chordwise coordinate $\eta \in [0,1]$ to $t = 2\eta - 1 \in [-1,1]$, the normalized chordwise basis functions are defined as:

$$Y_j(\eta) = \sqrt{\frac{2j+1}{2}} P_j(t), \qquad j = 0, \dots, n_c - 1, \tag{24}$$

where $P_j(t)$ denotes the standard Legendre polynomial of degree j(equivalent to the Jacobi polynomial $P_j^{(0,0)}(t)$). The chordwise functions satisfy exact $L_2$-orthonormality:

$$\int_0^1 Y_j(\eta) Y_l(\eta)\, d\eta = \delta_{jl}.$$

Spatial derivatives with respect to t are evaluated analytically using the Jacobi derivative identity:

$$\frac{d}{dt} P_j(t) = \frac{1}{2}(j+1) P_{j-1}^{(1,1)}(t).$$

The leading $(\eta = 0)$ and trailing $(\eta = 1)$ edges are physically free; the corresponding natural boundary conditions are satisfied variationally through the Rayleigh–Ritz energy minimization.

The combined $L_2$-orthonormality of both the spanwise $X_i(x)$ and chordwise $Y_j(\eta)$ basis families produces a global mass matrix that, for a flat plate of uniform thickness and orthogonal parametric lines, would be strictly diagonal. For the actual twisted, tapered airfoil shell, the mass matrix remains strongly diagonally dominant. This structural property underpins the well-conditioned mass-normalization procedure detailed in Section 3.6 and represents a decisive advantage over standard monomial or trigonometric Ritz bases, whose Gram matrices become severely ill-conditioned or numerically singular at moderate polynomial orders.

**Continuum Displacement Operators:** For the translational fields $(k = 1,2,3)$, the shape operator is $\mathbf{U}^{(k)} = \mathbf{e}_k \otimes \boldsymbol{\psi}$ with constant global Cartesian directions $\mathbf{e}_k$. For the rotational fields $(\alpha = 1,2)$, it is $\mathbf{U}^{(3+\alpha)} = \zeta \mathbf{e}_\alpha(x,\eta) \otimes \boldsymbol{\psi}$, whose parametric derivatives require the local frame derivatives $\mathbf{e}_{\alpha,x}$ and $\mathbf{e}_{\alpha,\eta}$ established in Section 2.5. The partial derivatives in Eq. (14) are evaluated at fixed $\eta$, consistent with the surface parametrization.

### 3.3. Constitutive Relations and Rotating Energy Functionals

3.3.1. Plane-Stress Elasticity and Transverse Shear Rigidity

For an isotropic, homogeneous material with Young's modulus E, Poisson's ratio $\nu$, and shear modulus $G = E/[2(1+\nu)]$, the plane-stress constitutive matrix $\mathbf{C}$, in the local frame is:

$$\mathbf{C} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 & 0 & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 & 0 & 0 \\ 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & k_s G & 0 \\ 0 & 0 & 0 & 0 & k_s G \end{bmatrix}, \tag{25}$$

with the shear correction factor $k_s = \pi^2/12$ [37, 61]. The transverse shear correction compensates for the constant through-thickness distribution of $\gamma_{13}, \gamma_{23}$ implied by Eq. (16).

3.3.2. Rotating Energy Functionals and System Matrices

Let $\mathbf{S}$ denote the skew-symmetric spin matrix corresponding to the angular velocity vector $\boldsymbol{\Omega} = \Omega \mathbf{e}_Y$, satisfying $\mathbf{S}\mathbf{a} = \boldsymbol{\Omega} \times \mathbf{a}$ for any vector $\mathbf{a}$:

$$\mathbf{S} = \begin{bmatrix} 0 & 0 & \Omega \\ 0 & 0 & 0 \\ -\Omega & 0 & 0 \end{bmatrix}, \qquad \mathbf{S}^T\mathbf{S} = -\mathbf{S}^2. \tag{26}$$

The absolute acceleration $\mathbf{a}$ of a material point observed within the rotating reference frame is expressed as:

$$\mathbf{a} = \ddot{\mathbf{u}} + 2\mathbf{S}\dot{\mathbf{u}} + \mathbf{S}\mathbf{S}(\mathbf{X} + \mathbf{u}) \tag{27}$$

comprising the relative acceleration $\ddot{\mathbf{u}}$, the Coriolis acceleration $2\mathbf{S}\dot{\mathbf{u}}$, and the centripetal acceleration $\mathbf{S}^2(\mathbf{X} + \mathbf{u})$.

Expressing the continuous displacement field in discretized form as $\mathbf{u}(x, \eta, \zeta, t) = \mathbf{U}(x, \eta, \zeta)\mathbf{q}(t)$, where $\mathbf{U}$ is the $3 \times n$ global displacement operator constructed from the Ritz basis polynomials and local director frame, application of D'Alembert's principle of virtual work:

$$\int_V \rho \ \delta\mathbf{u}^T\mathbf{a} \, dV + \int_V \delta\boldsymbol{\varepsilon}^T\boldsymbol{\sigma} \, dV = 0, \tag{28}$$

yields the semi-discrete non-linear governing equations of motion:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{G}\dot{\mathbf{q}} + (\mathbf{K}_{el} + \mathbf{K}_\sigma - \mathbf{K}_\Omega)\mathbf{q} = \mathbf{f}_c, \tag{29}$$

with the elastic stiffness matrix $\mathbf{K}_{el}$, mass matrix $\mathbf{M}$, gyroscopic matrix $\mathbf{G}$, spin-softening matrix $\mathbf{K}_\Omega$, and centrifugal body-force load vector $\mathbf{f}_c$ defined by the 3D volume integrals:

$$\begin{aligned} \mathbf{K}_{el} &= \int_V \mathbf{B}^T \mathbf{C} \mathbf{B} \, dV, & \mathbf{M} &= \int_V \rho \ \mathbf{U}^T\mathbf{U} \, dV, \\ \mathbf{G} &= 2\int_V \rho \ \mathbf{U}^T\mathbf{S}\mathbf{U} \, dV, & \mathbf{K}_\Omega &= \int_V \rho \ \mathbf{U}^T\mathbf{S}^T\mathbf{S}\mathbf{U} \, dV, \\ \mathbf{f}_c &= \int_V \rho \ \mathbf{U}^T\mathbf{S}^T\mathbf{S}\mathbf{X} \, dV, \end{aligned} \tag{30}$$

where the differential volume element $dV$ is mapped to the normalized parametric domain $\hat{\zeta} \in [-1, 1]$ as:

$$dV = \det \mathbf{J} \ \frac{h(x, \eta)}{2} \, d\hat{\zeta} \, d\eta \, dx \, . \tag{31}$$

The initial-stress geometric stiffness matrix $\mathbf{K}_\sigma$ depends on the static centrifugal stress field and is derived in detail in Section 3.5. The mass matrix $\mathbf{M}$, elastic stiffness matrix $\mathbf{K}_{el}$, and spin-softening matrix $\mathbf{K}_\Omega$ are inherently symmetric ($\mathbf{M}^T = \mathbf{M}$, $\mathbf{K}_{el}^T = \mathbf{K}_{el}$, $\mathbf{K}_\Omega^T = \mathbf{K}_\Omega$), whereas the gyroscopic matrix $\mathbf{G}$ is skew-symmetric ($\mathbf{G}^T = -\mathbf{G}$); these exact structural symmetries are enforced post-assembly to eliminate numerical round-off asymmetries. The negative sign of $\mathbf{K}_\Omega$ reflects the physical phenomenon of spin softening: the centripetal term $\mathbf{S}^2\mathbf{u} = -\mathbf{S}^T\mathbf{S}\mathbf{u}$ acts to reduce the effective dynamic restoring stiffness for structural displacements perpendicular to the spin axis. Furthermore, because the displacement operator $\mathbf{U}$ naturally incorporates the $\zeta$-weighted director rotation components from FSDT kinematics, both the translational and rotary inertia of the shell cross-section are captured consistently across $\mathbf{M}$, $\mathbf{G}$, and $\mathbf{K}_\Omega$, and the centrifugal body load vector $\mathbf{f}_c$ inherently includes the steady overturning moment generated by the through-thickness offset of the body force.

### 3.4. Singularity-Aware 3D Volume Quadrature

The volume integrands defining the mass, stiffness, and load matrices in Eqs. (30) and (31) (as well as the initial-stress geometric stiffness in Eq. (34)) are smooth functions along the spanwise coordinate $x$, so $n_q^{(x)}$ Gauss–Legendre quadrature points distributed over $[0, L]$ suffice for exact numerical integration. In the chordwise direction $\eta$, however, two distinct geometric features severely degrade the convergence rate of standard Gauss quadrature:

i. **Leading-edge thickness singularity:** The square-root behavior $\sqrt{\eta}$ in the NACA thickness law [Eq. (2)] enters the volume element $dV$ through $h(x,\eta)/2$ [Eq. (31)] and the stiffness/mass operators through hand $h^3$-type thickness moments, causing the chordwise thickness derivative to blow up at the leading edge ($\partial h_0/\partial\eta \to \infty$ as $\eta \to 0$).

ii. **Camber-curvature jump:** The second derivative of the camber line $f''(\eta)$ undergoes a finite jump discontinuity across the maximum-camber location $\eta = p$ [Eq. (1)]. This discontinuity propagates directly into $\mathbf{r}_{,\eta\eta}$ [Eq. (8)], the surface normal derivatives $\mathbf{n}_{,\eta}$ [Eq. (10)], the local frame derivatives $\mathbf{Q}_{,\eta}$ [Eq. (13)], and the 3D volume Jacobian determinant $\det \mathbf{J}$ [Eq. (15)].

To restore exponential (spectral) quadrature convergence, a two-part coordinate transformation and domain subdivision strategy is implemented:

**Leading-Edge Regularization Mapping ($\eta = t^2$):** The coordinate substitution $\eta = t^2$ (with $d\eta = 2t\,dt$) maps $\sqrt{\eta} \to t$, transforming the non-smooth square-root singularity into a smooth linear polynomial in $t \in [0, 1]$. The mapping Jacobian $2t$ is absorbed directly into the chordwise quadrature weights $(w_\eta = 2t\,w_t)$. Standard Gauss–Legendre quadrature in tthen converges exponentially.

**Camber-Curvature Domain Subdivision:** Because $\eta = t^2$, the maximum-camber location $\eta = p$ corresponds to $t = \sqrt{p}$. The integration interval $t \in [0, 1]$ is divided at $t = \sqrt{p}$ into two sub-intervals, $[0, \sqrt{p}\,]$ and $[\sqrt{p}, 1]$, so that the camber line within each sub-interval is polynomially smooth. On each sub-interval, $n_q^{(\eta)}$Gauss–Legendre points are placed independently. For symmetric profiles $(m = 0)$, no split is required and a single integration domain $t \in [0, 1]$ is used.

**Through-Thickness Layered Quadrature:** Through the thickness, $n_q^{(\zeta)}$ Gauss–Legendre points are placed on $\zeta \in [-1, 1]$, where physical thickness coordinates are mapped via $z = \zeta\, h(x,\eta)/2$. The integrands in $\zeta$ are at most quadratic polynomials multiplied by the rational function $\det \mathbf{J}(\zeta)$ [Eq. (15)]. Setting $n_q^{(\zeta)} \geq 2$ (typically $n_q^{(\zeta)} = 3$) guarantees that both translational and rotary inertia terms are captured with high precision.

The total number of 3D volume quadrature points is $n_q^{(x)} \times n_q^{(\eta)} \times n_{sub} \times n_q^{(\zeta)}$ (where $n_{sub} = 2$ for cambered profiles and 1 for symmetric profiles). To prevent numerical rank deficiency in the discretized global matrices, the quadrature orders satisfy $n_q^{(x)} \geq n_s + 1$ and $n_q^{(\eta)} \geq n_c + 1$.

### 3.5. Prestress-Consistent Geometric Stiffness and Stress Recovery

Centrifugal stiffening arises from the interaction of the steady stress field with the second-order (rotation) part of the strain. Rather than adopting a beam-type axial-force approximation or a membrane-resultant approximation, the present formulation evaluates the complete three-dimensional stress state produced by the centrifugal load and integrates it against the full displacement gradient.

**Static equilibrium.** The steady deformation $\mathbf{q}_0$ on the reference geometry is obtained from:

$$\mathbf{K}_{el}\,\mathbf{q}_0 = \mathbf{f}_c, \tag{32}$$

solved in mass-normalized coordinates after verification that the balanced stiffness admits a Cholesky factorization; failure indicates that the reference configuration is not a stable static state and that a nonlinear equilibrium path would be required. The relative residual $(\|\mathbf{K}_{el}\,\mathbf{q}_0 - \mathbf{f}_c\|)/\|\mathbf{f}_c\|$ is recorded as a solution diagnostic.

**Stress recovery.** At every volume quadrature point, the 5-component strain vector $\boldsymbol{\varepsilon}_0 = \mathbf{B}\mathbf{q}_0$ and Cauchy stress vector $\boldsymbol{\sigma}_0 = \mathbf{C}\boldsymbol{\varepsilon}_0$ are evaluated and assembled into the symmetric local stress tensor $\boldsymbol{\Sigma}_0$:

$$\boldsymbol{\Sigma}_0 = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & 0 \end{bmatrix}. \tag{33}$$

**Initial-stress stiffness.** Linearizing the Green–Lagrange strain about the prestressed state and retaining the quadratic term yields the classical initial-stress operator [41,45,62]:

$$\delta W_\sigma = \int_V \sigma^0_{ab}\ \delta u_{k,a}\, u_{k,b}\, dV \quad \Rightarrow \quad \mathbf{K}_\sigma = \int_V \sum_{k=1}^{3} (\nabla_\ell U_k)^\top \boldsymbol{\Sigma}_0 (\nabla_\ell U_k) dV, \tag{34}$$

where $\nabla_l U_k$ is the row of the operator giving the derivative of the k-th Cartesian displacement component along the three local directions. All nine gradient components and all five independent stress components participate; no resultant or beam approximation is introduced. This is the sense in which the formulation is prestress-consistent: the same kinematic operator, the same quadrature, and the same three-dimensional Jacobian are used for the static solution, the stress recovery, and the geometric stiffness.

**Validity of the linear reference-state approximation.** Equation (32) is a linear solve on the undeformed geometry[4]. Its adequacy is checked a posteriori by three quantities reported for every analysis: the maximum linear strain $\max|\boldsymbol{\varepsilon}_0|$(threshold set at 1%), the maximum static mid-surface displacement normalized by span (threshold set at 2%), and the maximum director slope $\sqrt{\phi_1^2 + \phi_2^2}$ (threshold set at 0.05 rad). Exceedance of any threshold indicates that a geometrically non-linear equilibrium and tangent operator would be required.

### 3.6. Mass Normalization and Energy-Scaled Gyroscopic Eigensolver

Setting $\mathbf{f}_c$ aside (it determines the static prestress deformation $\mathbf{q}_0$ only) and substituting $\mathbf{q}(t) = \boldsymbol{\phi} e^{\lambda t}$ into the homogeneous form of Eq. (29) yields the quadratic eigenvalue problem:

$$(\lambda^2 \mathbf{M} + \lambda\, \mathbf{G} + \mathbf{K})\boldsymbol{\phi} = \mathbf{0}, \qquad \mathbf{K} = \mathbf{K}_{el} + \mathbf{K}_\sigma - \mathbf{K}_\Omega. \tag{35}$$

where $\lambda \in \mathbb{C}$ is the complex eigenvalue, $\boldsymbol{\phi} \in \mathbb{C}^n$ is the corresponding eigenvector (mode shape), and $\mathbf{K}$ represents the combined effective stiffness matrix.

The natural cyclic frequency of a mode is $f = \mathrm{Im}(\lambda)/2\pi$ (in Hz) and its growth/decay rate is given by $\mathrm{Re}(\lambda)$. For $K \succ 0$ (strictly positive-definite total stiffness matrix), the governing system behaves as a conservative gyroscopic system whose eigenvalues are purely imaginary ($\mathrm{Re}(\lambda) = 0$) and whose motion is dynamically stable [55, 56, 63, 64]. Loss of positive-definiteness of K(which spin softening $K_\Omega$ can produce at sufficiently high rotational speeds) admits eigenvalues with a positive real part ($\mathrm{Re}(\lambda) > 0$), corresponding to divergence-type dynamic instability. Detecting and quantifying this stability boundary is a primary objective of the present solver.

To preserve numerical precision across the disparate scales of translational and ζ-weighted rotational degrees of freedom, the mass matrix is diagonally equilibrated, $\widetilde{\mathbf{M}} = \mathbf{DMD}$ with $\mathbf{D} = \mathrm{diag}\left(M_{ii}^{-1/2}\right)$, and factorized via Cholesky decomposition, $\widetilde{\mathbf{M}} = \mathbf{LL}^T$. The congruence transformation $\mathbf{q} = \mathbf{Ty}$ with $\mathbf{T} = \mathbf{DL}^{-T}$ reduces the quadratic gyroscopic eigenvalue problem to standard mass-orthonormal form:

$$\left(\lambda^2 \mathbf{I} + \lambda \widetilde{\mathbf{G}} + \widetilde{\mathbf{K}}\right)\mathbf{y} = 0,$$

where $\widetilde{\mathbf{K}} = \mathbf{T}^T\mathbf{KT}$ is symmetric and $\widetilde{\mathbf{G}} = \mathbf{T}^T\mathbf{GT}$ is real skew-symmetric.

When the effective stiffness is positive-definite ($\lambda_{min}\left(\widetilde{\mathbf{K}}\right) > 0$), the spectral decomposition $\widetilde{\mathbf{K}} = \mathbf{V\Lambda V}^T$ allows scaling the state vector by the square-root stiffness, $\mathbf{z} = \left\{\boldsymbol{\Lambda}^{1/2}\mathbf{; V}^T\mathbf{y}\ \ \mathbf{V}^T\dot{\mathbf{y}}\right\}$. The first-order state equation becomes:

$$\dot{\mathbf{z}} = \mathbf{Az},\ \mathbf{A} = \begin{bmatrix} \mathbf{0} & \boldsymbol{\Lambda}^{1/2} \\ -\boldsymbol{\Lambda}^{1/2} & -\mathbf{V}^T\widetilde{\mathbf{G}}\mathbf{V} \end{bmatrix},$$

where Ais strictly real skew-symmetric ($\mathbf{A}^T = -\mathbf{A}$). Consequently, i$\mathbf{A}$ is Hermitian, ensuring that a standard LAPACK real symmetric eigensolver (scipy.linalg.eigh) returns purely imaginary complex eigenvalues ($\lambda = -i\mu$) with zero spurious real parts. Physical modal vectors are recovered via $\boldsymbol{\phi} = \mathbf{TV\Lambda}^{-1/2}\mathbf{z}_{(1)}$.

If spin-softening causes stiffness loss of positive-definiteness ($\lambda_{\min}(\widetilde{\mathbf{K}}) \leq 0$), the solver transitions to a time-scaled $2n \times 2n$ companion state formulation solved via a general non-symmetric LAPACK eigensolver (scipy.linalg.eig), where positive real parts ($\mathrm{Re}(\lambda) > 10^{-7}$) identify growing divergence-type instabilities. All computed eigenpairs $(\lambda, \boldsymbol{\phi})$ are normalized to unit modal mass and verified by evaluating relative residuals ($r \leq 10^{-6}$) on the unconstrained global matrices.

## 4. Computational and Numerical Implementation

### 4.1. One-Pass Matrix Assembly and Volume Check

All linear system operators defining the unconstrained dynamic system are assembled in a single vectorized pass over the 3D volume quadrature points. To strictly bound peak memory usage, quadrature points are evaluated layer-by-layer along the through-thickness coordinate $\zeta \in [-1, 1]$. At each layer, physical coordinates $z = \zeta h(x, \eta)/2$ are assigned across surface quadrature points $(x, \eta)$. The displacement shape operator $\mathbf{U}$, local gradient $\nabla_l \mathbf{U}$, strain-displacement matrix $\mathbf{B}$, Jacobian determinant $\det \mathbf{J}$ [Eq. (15)], and material position $\mathbf{X}$ [Eq. (14)] are evaluated in vectorized form. The elemental contributions

$$\mathbf{B}^\top \mathbf{C} \mathbf{B}, \quad \rho\, \mathbf{U}^\top \mathbf{U}, \quad 2\rho\, \mathbf{U}^\top \mathbf{S} \mathbf{U}, \quad \rho\, (\mathbf{S}\mathbf{U})^\top (\mathbf{S}\mathbf{U}), \quad \rho\, \mathbf{U}^\top \mathbf{S}^\top \mathbf{S} \mathbf{X}$$

are accumulated into global system matrices using effective weights $w = w_x w_\eta w_\zeta\, h(x, \eta)/2 \det \mathbf{J}$, yielding $\mathbf{K}_{\mathrm{el}}$, $\mathbf{M}$, $\mathbf{G}$, $\mathbf{K}_\Omega$, and $\mathbf{f}_c$ [Eq. (30)]. Simultaneously, physical blade volume $V = \int \mathrm{d}V$ and mass $m = \rho V$ are accumulated and verified against the CAD model to confirm mapping accuracy and Jacobian positivity ($\det \mathbf{J} > 0$). Following the static solve $\mathbf{q}_0$ [Eq. (32)], a second vectorized pass evaluates Cauchy stress $\boldsymbol{\Sigma}_0$ [Eq. (33)] and assembles geometric stiffness $\mathbf{K}_\sigma$ [Eq. (34)].

### 4.2. Software Stack and Reproducibility

The formulation is implemented in Python using NumPy [70] and SciPy [71] (utilizing LAPACK-backed dense linear algebra [68] for Jacobi and Legendre polynomial evaluation) with Matplotlib [72] for post-processing. For each numerical simulation, the code exports the modal frequency table with growth rates and relative residuals, the complete complex eigen spectrum, the assembled global system operators, the static coefficient vector, and a JSON record summarizing all configuration parameters and numerical diagnostics. Root and tip section point clouds are exported in a CAD-compatible tabular format to permit the construction of a geometrically identical 3D solid model in commercial software for finite element validation.

### 4.3. Baseline Reference Blade Configuration and CAD Export

To evaluate the convergence, numerical accuracy, and dynamic characteristics of the proposed semi-analytical framework, a baseline cantilevered compressor blade with a pre-twisted, tapered, cambered NACA 6412 profile is defined. The baseline geometry, material properties, rotational operating parameters, and numerical discretization orders used across the benchmark studies in Section 5 are summarized in Table 1 below.

Table 1. Reference blade physical parameters, material properties, and Ritz discretization settings.

| Category | Quantity | Symbol | Value |
|---|---|---|---|
| Blade Geometry | Span length | L | 0.3 m |
| | Root chord | $c_0$ | 0.2 m |
| | Cross-section profile | — | NACA 6412 |
| | Chord taper fraction | $\tau_c$ | 0.10 |
| | Thickness taper fraction | $\tau_h$ | 0.50 |
| | Root stagger angle | $\theta_0$ | $-20°$ |
| | Total tip pre-twist | $\theta_{tip}$ | $-15°$ |
| | Pre-twist axis chord fraction | $\eta_a$ | 0.25(quarter-chord) |
| | Rotor hub radius | $R_h$ | 0.2 m |
| Operational State | Rotor speed | N ($\Omega$) | 4000 rpm |
| Material Properties | Young's modulus | E | 200 GPa |
| | Poisson's ratio | $\nu$ | 0.30 |
| | Density | $\rho$ | 7850 kg $m^{-3}$ |

| | | | |
|---|---|---|---|
| | Shear correction factor | $\kappa$ | $\pi^2/12 \approx 0.8225$ |
| Ritz Discretization | Ritz polynomial order | $n_s \times n_c$ | $6 \times 6$ (180 DOFs) |
| | Volume quadrature points | $n_q^{(x)}, n_q^{(\eta)}, n_q^{(\zeta)}$ | $10 \times (10 \times 2) \times 3 = 600$points |

## 5. Results and Discussion

### 5.1. Geometric Verification and Volume Integration Accuracy

Prior to dynamic and modal analyses, the analytical geometric mapping derived in Section 2 was verified against the 3D CAD solid model constructed in ANSYS. To ensure complete geometric consistency between the semi-analytical solver and the finite element baseline, 12 spanwise cross-sectional station coordinates evaluated in the Python solver were exported to .txt file and imported into ANSYS to loft the 3D CAD solid model (Figure 4). Figure 2 illustrates the cross-sectional profiles generated analytically at the blade root ($x = 0$) and tip ($x = L$). The plot displays the upper (suction) and lower (pressure) skin boundaries, the camber line, the straight chord line, and the pre-twist stacking axis located at the quarter-chord point ($\eta_a = 0.25$). The tip section demonstrates the combined spanwise tapering and pre-twist functions: a 10% chord reduction ($\tau_c = 0.10$), a 50% thickness taper ($\tau_h = 0.50$), and a relative tip pre-twist of $-15°$($\theta_{tip} = -15°$) relative to the $-20°$root stagger angle ($\theta_0 = -20°$).

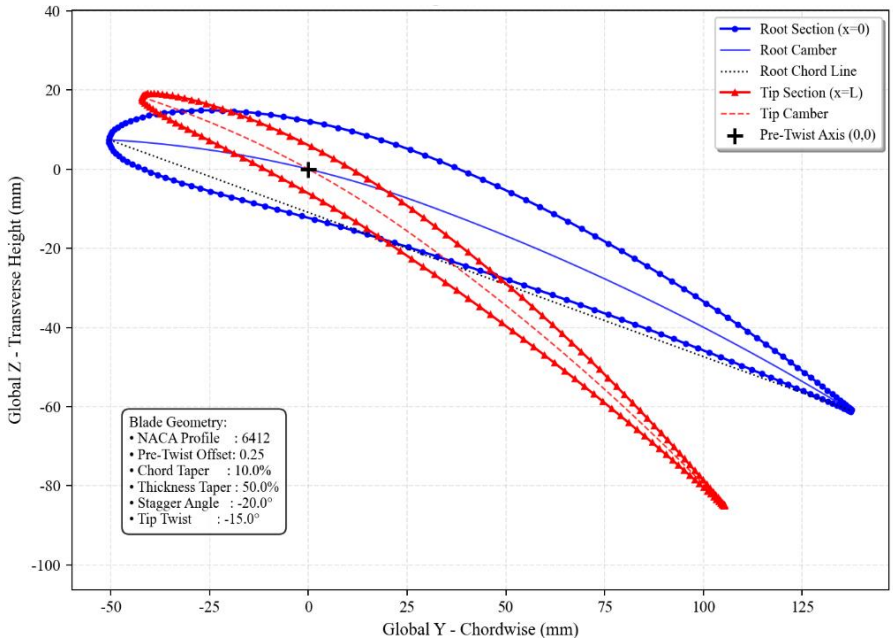



Figure 2. Root and tip section geometry: Overlay of the analytically generated cross-sections at the blade root ($x = 0$, solid curve) and tip ($x = L$, dashed curve) in the local section plane (Y–Z), showing upper and lower skin envelopes, camber lines, straight chord lines, and the quarter-chord stacking axis at $(0, 0)$. The tip profile illustrates the 10% chord reduction, 50% thickness taper, and $-15°$relative pre-twist.

The continuous 3D surface mapping generated by the analytical formulation is compared against the reconstructed solid CAD model in Figure 3 and Figure 4. Figure 3 shows the analytical 3D representation of the blade, illustrating

the smooth, continuous variation of pre-twist, taper, and camber along the span. Figure 4 displays the corresponding 3D solid CAD model created in ANSYS by lofting through the exported station point clouds.

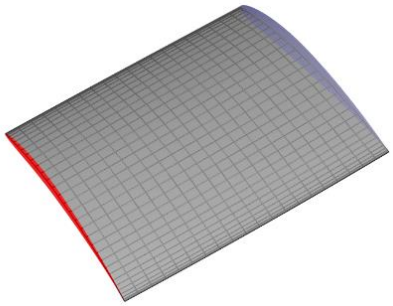


Figure 3. Analytical 3D blade representation

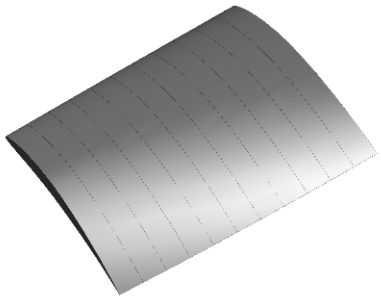


Figure 4. ANSYS CAD model reconstruction

To verify 3D volume integration and the singularity-aware quadrature scheme (Section 3.4), the total integrated blade volume ($V = \int_V \det J_r \, dV$) and mass ($m = \rho V$) were compared against the ANSYS CAD solid model report (Table 2). The Rayleigh–Ritz volume matches the CAD model to within $0.221\%$ relative difference.

Table 2. Comparison of integrated blade volume and mass between the present Rayleigh–Ritz formulation and the 3D ANSYS CAD solid model for the reference NACA 6412 blade.

| Parameter | ANSYS CAD Model | Present Rayleigh–Ritz | Relative Difference (%) |
|---|---|---|---|
| Mass (kg) | 5.36798535 | 5.35611030 | -0.2212% |
| Volume ($m^3$) | 0.00068382 | 0.00068231 | -0.2208% |

As listed in Table 2, the Rayleigh–Ritz volume matches the ANSYS CAD solid model to within 0.2208% relative difference, and the integrated mass matches to within 0.2212%. This close agreement confirms the accuracy of the exact 3D volume metric tensor and First-Order Shear Deformation Theory (FSDT) integration over complex pre-twisted, tapered geometries.

### 5.2. Convergence Characteristics of Orthonormal Ritz Polynomials

To ensure numerical stability and determine the optimal expansion order for spectral accuracy, a systematic convergence study was performed by varying the Ritz polynomial order $n_s = n_c$ from $4 \times 4$ to $7 \times 7$ (corresponding to system DOFs $N = 80$ to $N = 245$). As shown in Table 3, the natural frequencies converge rapidly and monotonically toward the 3D ANSYS finite element baseline.

Table 3. Convergence of natural frequencies (Hz) at N = 4,000 rpm for increasing Ritz order ($n_s \times n_c$) compared against ANSYS FEA.

| Ritz Order | Model / Error | Mode 1 | Mode 2 | Mode 3 | Mode 4 | Mode 5 | Mode 6 | Mode 7 |
|---|---|---|---|---|---|---|---|---|
| 4×4 | Present (RR) | 248.87 | 724.04 | 1005.19 | 1623.39 | 1734.83 | 2464.40 | 2906.82 |
| | ANSYS FEA | 245.85 | 706.85 | 970.04 | 1606.69 | 1659.01 | 2098.47 | 2419.68 |
| | Error (%) | 1.23% | 2.43% | 3.62% | 1.04% | 4.57% | 17.44% | 20.13% |
| 5×5 | Present (RR) | 246.86 | 712.70 | 972.58 | 1609.34 | 1664.66 | 2226.00 | 2446.49 |
| | ANSYS FEA | 245.85 | 706.85 | 970.04 | 1606.69 | 1659.01 | 2098.47 | 2419.68 |
| | Error (%) | 0.41% | 0.83% | 0.26% | 0.17% | 0.34% | 6.08% | 1.11% |
| 6×6 | Present (RR) | 246.01 | 708.17 | 969.46 | 1606.47 | 1657.86 | 2116.92 | 2413.46 |
| | ANSYS FEA | 245.85 | 706.85 | 970.04 | 1606.69 | 1659.01 | 2098.47 | 2419.68 |
| | Error (%) | 0.07% | 0.19% | -0.06% | -0.01% | -0.07% | 0.88% | -0.26% |
| 7×7 | Present (RR) | 245.66 | 706.49 | 968.41 | 1605.21 | 1655.06 | 2103.02 | 2405.94 |
| | ANSYS FEA | 245.85 | 706.85 | 970.04 | 1606.69 | 1659.01 | 2098.47 | 2419.68 |
| | Error (%) | -0.08% | -0.05% | -0.17% | -0.09% | -0.24% | 0.22% | -0.57% |

At the lowest expansion order of $4 \times 4$ (80 DOFs), relative errors with respect to ANSYS range from 1.04% to 20.13% for the lowest seven modes. Increasing the order to $5 \times 5$(125 DOFs) markedly improves accuracy, reducing all relative errors below 6.08%. At $6 \times 6$(180 DOFs), the present Rayleigh–Ritz formulation achieves essentially complete agreement with ANSYS, with a maximum error of 0.07% for Mode 1 and errors below 0.88% for all primary modes. Further increasing the order to $7 \times 7$(245 DOFs) produces negligible frequency changes. Accordingly, an expansion order of $n_s = n_c = 6$($N = 180$ DOFs) offers an optimal balance between spectral accuracy and computational efficiency and is therefore adopted for all subsequent rotating benchmark studies.

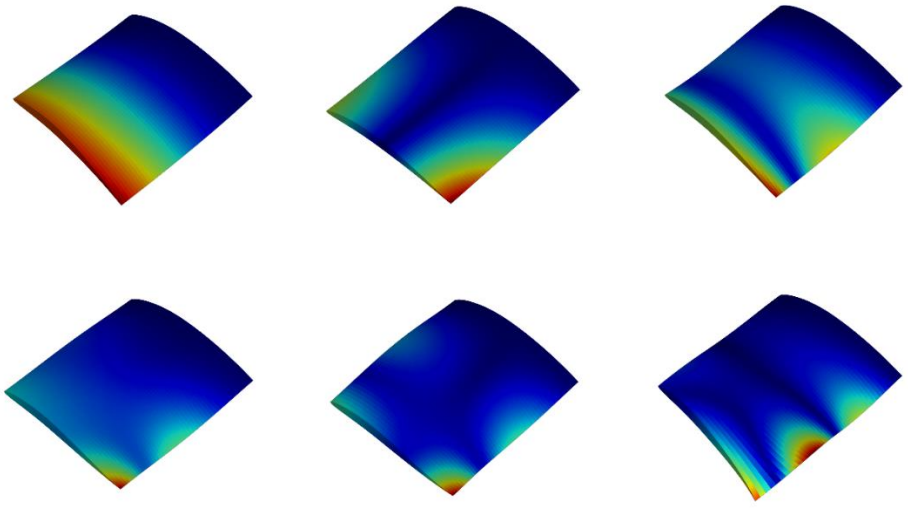
### 5.3. Non-Rotating Modal Validation against 3D Solid FEA (0 RPM)

**Table 4.** Non-rotating frequency comparison with ANSYS.

| Modes | ANSYS | RR | Error |
|---|---|---|---|
| 1 | 231.0974 | 231.301 | 0.09% |
| 2 | 701.4673 | 702.82 | 0.19% |
| 3 | 950.0262 | 949.296 | -0.08% |
| 4 | 1601.545 | 1601.127 | -0.03% |
| 5 | 1647.969 | 1646.059 | -0.12% |
| 6 | 2075.323 | 2094.392 | 0.92% |
| 7 | 2415.01 | 2408.762 | -0.26% |

(RPM: 0.01 rpm, used here)

Caption: Lowest seven natural frequencies (Hz) of the reference blade at $N = 0$: present method (150 DOF) versus ANSYS Mechanical using (a) quadratic shell elements (SHELL281) on the mid-surface with the exact thickness distribution applied as a field, and (b) quadratic solid elements (SOLID186) on the CAD solid. State the element counts and degrees of freedom of each FE model, the mesh-convergence tolerance achieved, and the percentage difference of the present method from each. Discuss the residual differences in terms of the plane-stress assumption, the shear correction factor, and the finite trailing-edge face modelled in the solid.



(From code-RR)

**Figure 5 — Non-rotating mode shapes: present versus ANSYS.** Caption: Side-by-side deformed skins for the lowest six modes, colored by displacement magnitude on a common scale. Identify each mode by its dominant character (first flap, first edgewise, first torsion, second flap, chordwise bending, coupled flap–torsion) and comment on the flap–torsion coupling induced by pre-twist and the stacking-axis offset from the section centroid. Report the Modal Assurance Criterion between corresponding present and FE mode shapes evaluated on a common grid of mid-surface points.

### 5.4. Rotating Modal Validation and Prestress Analysis (4,000 RPM and 20,000 RPM)

**[PLACEHOLDER: Table 5 — Static prestress diagnostics.]** Caption: For N ∈ {1000, ... ,8000} rpm: static solve relative residual, maximum linear strain, maximum mid-surface displacement normalised by L, maximum director slope, and the resulting maximum von Mises stress at the root. Compare the root radial stress and tip displacement with the ANSYS static prestress solution (rotational inertia load). Mark the speed, if any, at which a validity threshold of is exceeded.

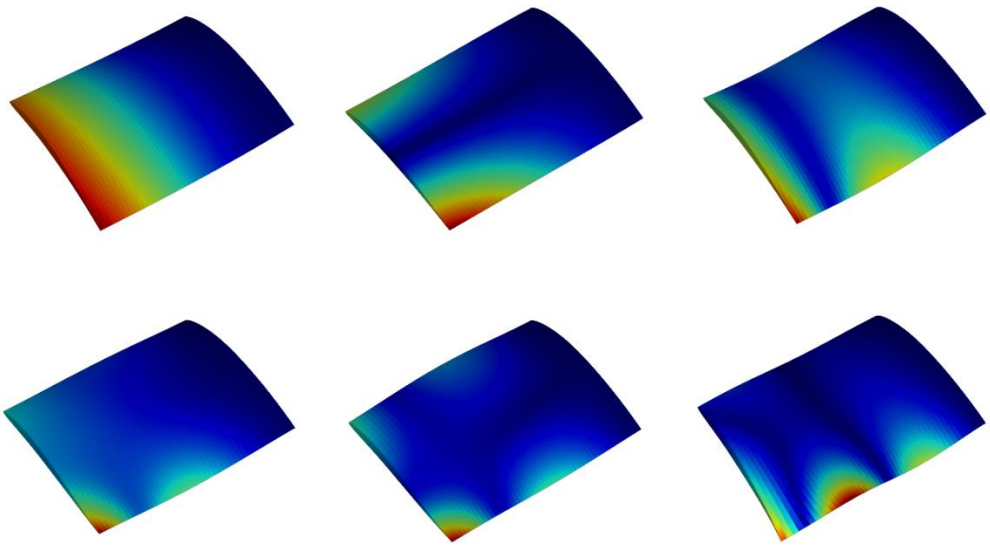
**[PLACEHOLDER: Figure 6 — Centrifugal stress field.]** Caption: Contours of the mid-surface radial stress $\sigma_{11}$ and shear stress $\sigma_{12}$ recovered from $\mathbf{q}_0$ at 4000 rpm, showing the concentration toward the root and the chordwise redistribution induced by pre-twist and stagger. Compare with the corresponding ANSYS contour.

**Table 6 — Rotating frequency comparison with ANSYS**: Lowest seven frequencies at 4000 rpm: present method (with $\mathbf{K}_\sigma$, $\mathbf{K}_\Omega$, and $\mathbf{G}$) versus ANSYS prestressed modal analysis in the rotating frame with Coriolis effects activated and spin softening included. Provide percentage differences and identify any modal reordering with respect to the non-rotating case.

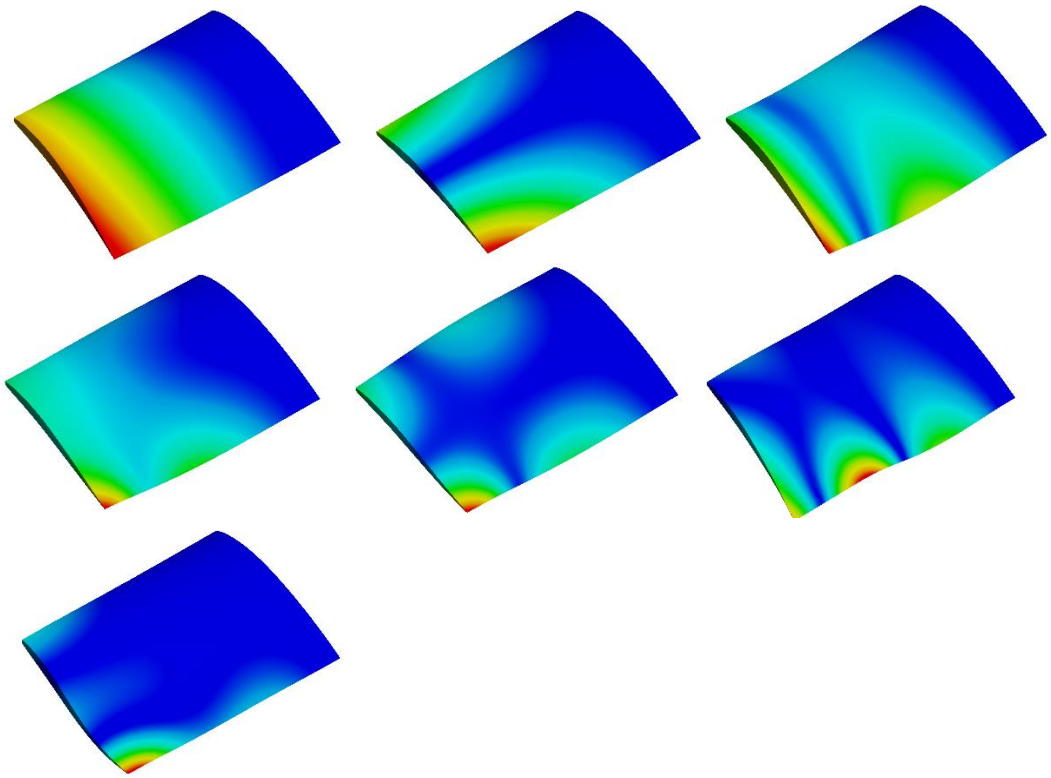
| Modes | ANSYS | Present (RR) | Error |
|---|---|---|---|
| 1 | 245.85 | 246.012 | 0.07% |
| 2 | 706.85 | 708.174 | 0.19% |
| 3 | 970.04 | 969.46 | -0.06% |
| 4 | 1606.69 | 1606.47 | -0.01% |
| 5 | 1659.01 | 1657.864 | -0.07% |
| 6 | 2098.47 | 2116.923 | 0.88% |
| 7 | 2419.68 | 2413.462 | -0.26% |

Mode Shapes

**From Code**: First six modes:



**From ANSYS:** First seven modes:



Another table:

| Modes | RPM: 20000 | Pre-twist: 15° | Stagger: 20° |
|---|---|---|---|
| | ANSYS (Hz) | RR | R_hub: 0.2m |
| | | $n_s \times n_c$: 6×6 | % Difference |
| 1 | 464.7071 | 462.259 | -0.53% |
| 2 | 822.2366 | 821.832 | -0.05% |
| 3 | 1343.871 | 1346.649 | 0.21% |
| 4 | 1705.789 | 1720.101 | 0.84% |
| 5 | 1928.945 | 1934.76 | 0.30% |
| 6 | 2453.493 | 2447.106 | -0.26% |
| 7 | 2646.897 | 2660.499 | 0.51% |

### 5.5. Campbell Diagram and Speed-Dependent Behavior

**[PLACEHOLDER: Figure 7 — Campbell diagram.]** Caption: Natural frequencies of the lowest eight modes versus rotor speed from 0 to a maximum speed exceeding the operating range, with engine-order lines 1E to 6E. Identify the crossings of the first flap and first torsion modes with low engine orders, and mark the operating speed. Overlay the ANSYS results at discrete speeds as symbols. Discuss the different sensitivities of flap-dominated modes (strong stiffening) versus torsion- and edgewise-dominated modes (weak stiffening or softening) and relate them to the stress components entering $\boldsymbol{K}_\sigma$.

**[PLACEHOLDER: Figure 8 — Decomposition of rotational effects.]** Caption: Frequency shift of the first three modes versus speed with (a) $\boldsymbol{K}_\sigma$ only, (b) $\boldsymbol{K}_\sigma - \boldsymbol{K}_\Omega$, (c) $\boldsymbol{K}_\sigma - \boldsymbol{K}_\Omega$ with $\boldsymbol{G}$. Quantify the relative magnitudes of centrifugal stiffening, spin softening, and Coriolis coupling for a blade of this aspect ratio and stagger, and identify the speed at which the Coriolis-induced splitting of closely spaced modes becomes discernible.

**[PLACEHOLDER: Figure 9 — Complex mode shapes and Coriolis coupling.]** Caption: Real and imaginary parts of the mid-surface displacement for a pair of Coriolis-coupled modes at the operating speed, showing the phase lag

between flap and edgewise components that distinguishes a gyroscopic (travelling) mode from a standing mode. Report the modal phase collinearity index.

### 5.6. Dynamic Stability and Divergence Boundaries

**[PLACEHOLDER: Figure 10 — Stability map.]** Caption: Minimum eigenvalue $\lambda_{min}(\tilde{K})$ of the mass-normalised stiffened operator and maximum growth rate $maxRe\lambda$ versus rotor speed, extending to speeds at which $\lambda_{min}$ becomes negative and the solver switches to the general branch. Identify the divergence speed and compare it with the speed at which the ANSYS prestressed modal analysis reports a zero or negative frequency. Note that, because the present model is linear about the reference geometry, the static-validity thresholds will typically be exceeded before divergence; discuss the implications for interpreting the predicted divergence speed as an upper bound on the linear-model validity rather than as a physical limit.

**[PLACEHOLDER: Table 7 — Growing modes.]** Caption: At the first speed for which unstable eigenvalues are detected: list each growing eigenvalue (real part in $s^{-1}$, imaginary part in Hz), its residual, and the dominant character of its mode shape.

### 5.7. Comparative Benchmark Validation against Published Literature

Li-Cheng paper [101]

This is simplified/simple plate

Constant thickness(3mm) rectangular plate.

| | Pre twist: 15° | offset: 0.5 | RPM: 4k | Stagger: 45° | |
|---|---|---|---|---|---|
| Table 3 | My RR | Li-Cheng Paper ANSYS | % Dev | Li-Cheng Paper | % Dev |
| | 7.496 | 7.4951 | 0.01% | 7.5559 | -0.79% |
| | 19.470 | 19.3556 | 0.59% | 19.4017 | 0.35% |
| | 23.116 | 23.0676 | 0.21% | 23.298 | -0.78% |
| | 39.486 | 39.3061 | 0.46% | 40.494 | -2.49% |
| | 45.890 | 45.6935 | 0.43% | 45.5882 | 0.66% |
| | | RMSE | 0.397% | | 1.265% |
| | | | | | |
| | My RR | My ANSYS | % Dev | | |
| | 91.116 | 91.091 | 0.03% | | |
| | 236.654 | 236.481 | 0.07% | | |
| | 280.977 | 280.486 | 0.18% | | |
| | 479.962 | 472.819 | 1.51% | | |
| | 557.793 | 554.120 | 0.66% | | |
| | 629.034 | 621.176 | 1.26% | | |
| | 820.376 | 805.750 | 1.82% | | |
| | | | | | |
| Parameter | ANSYS | Code-RR | % Diff | | |
| Mass | 1.42291519 | 1.41479059 | -0.5710% | | |
| Volume | 0.00018126 | 0.00018023 | -0.5699% | | |

| **RPM: 8k** | **offset: 0.5** | **Stagger: 45°** | | |
|---|---|---|---|---|
| **My RR** | **Li-Cheng Paper ANSYS** | **% Dev** | **Li-Cheng Paper** | **% Dev** |

| | | | | |
|---|---|---|---|---|
| 14.145 | 14.1175 | 0.19% | 14.3692 | -1.56% |
| 25.406 | 25.2388 | 0.66% | 25.2255 | 0.72% |
| 38.400 | 38.2925 | 0.28% | 39.2559 | -2.18% |
| 45.417 | 45.2517 | 0.37% | 46.3747 | -2.07% |
| 58.239 | 57.9657 | 0.47% | 56.874 | 2.40% |
| | RMSE | 0.427% | | 1.883% |
| | | | | |
| **My RR** | **My ANSYS** | **% Dev** | | |
| 171.939 | 171.682 | 0.15% | | |
| 308.811 | 308.025 | 0.26% | | |
| 466.759 | 464.781 | 0.43% | | |
| 552.046 | 545.985 | 1.11% | | |
| 707.898 | 703.099 | 0.68% | | |
| 871.709 | 865.160 | 0.76% | | |
| 969.147 | 953.755 | 1.61% | | |
| | RMSE | 0.86% | | |

### 5.8. Parametric Sensitivity Studies (Pre-twist, Taper, Camber, Hub Radius, Stagger)

**[PLACEHOLDER: Figure 11 — Effect of pre-twist.]** Caption: Lowest six frequencies at 0 and 4000 rpm versus total pre-twist $\theta_{tip} \in [-45°, 0°]$. Show the increasing flap–torsion coupling and the frequency veering between the second flap and first torsion branches.

**[PLACEHOLDER: Figure 12 — Effect of taper and camber.]** Caption: Frequencies versus (a) thickness taper $\tau_h \in [0,0.7]$, (b) chord taper $\tau_c \in [0,0.4]$, and (c) camber, comparing NACA 0012, 2412, 4412, and 6412 sections at fixed thickness. Discuss the opposing influence of camber on chordwise bending stiffness and torsional inertia.

**[PLACEHOLDER: Figure 13 — Effect of hub radius and stagger.]** Caption: Rotating frequencies at 4000 rpm versus hub radius $R_h/L \in [0,2]$ and root stagger $\theta_0 \in [-60°, 0°]$. Relate the hub-radius sensitivity to the radial stress level and the stagger sensitivity to the redistribution of the centrifugal load between the flap and edgewise directions and to the in-plane component of $\boldsymbol{K}_{\Omega}$.

### 5.9. 5.9 Computational Efficiency and DOF Reduction

**[PLACEHOLDER: Table 8 — Computational effort.]** Caption: Wall-clock time (single BLAS thread, specified CPU), peak memory, and degree-of-freedom count for the present method at Ritz orders 4×4, 6×6, 8×8, and 10×10, against the ANSYS shell and solid models of Tables 4 and 6 (including meshing, static prestress, and modal solve). Highlight the ratio of degrees of freedom required to achieve 0.5 % agreement on the first eight rotating frequencies.

## 6. Conclusions

A semi-analytical formulation has been developed for the free vibration and dynamic stability of rotating compressor blades represented as pre-twisted, tapered, cambered NACA four-digit airfoil shells. The formulation combines two elements that have hitherto been confined to separate methodological traditions. From the degenerated-continuum shell concept it takes the exact three-dimensional Jacobian mapping, so that the strain field is obtained from the true Cartesian displacement gradient without recourse to the Lamé parameters and curvature approximations of classical shell theories; from the Rayleigh–Ritz tradition it takes a global, $L_2$-orthonormal polynomial basis—scaled Jacobi functions in the span that enforce the clamped root exactly, and normalised Legendre functions in the chord—so that the entire blade is represented by a few hundred degrees of freedom without a finite element mesh.

The rotational physics is treated consistently in the blade-fixed frame. The centrifugal body force is integrated to a consistent load, the three-dimensional static equilibrium is solved, and the complete recovered stress tensor drives the initial-stress stiffness through the same kinematic operator and quadrature used for the dynamic operators. Coriolis coupling and spin softening enter through a skew-symmetric gyroscopic matrix and a centrifugal stiffness matrix that

include rotary inertia. A singularity-aware chordwise quadrature restores spectral integration accuracy in the presence of the square-root leading-edge behavior of the NACA thickness law and the curvature discontinuity of the camber line.

The quadratic gyroscopic eigenvalue problem is solved by an energy-scaled reformulation whose state matrix is exactly real skew-symmetric when the stiffened operator is positive definite. The Hermitian eigenproblem that results returns purely imaginary eigenvalues, so that natural frequencies are free of the spurious real parts that contaminate generalized solvers on ill-scaled companion matrices and that would otherwise be misinterpreted as damping or growth. When positive-definiteness is lost, a time-scaled general solve preserves the growth rates that identify divergence. Every static and dynamic solution is accompanied by residual and validity diagnostics, permitting the range of applicability of the linear reference-state prestress approximation to be delimited objectively.

The verification and validation program of Section 5 establishes the convergence of the Ritz expansion and quadrature, the agreement with shell- and solid-element finite element solutions at rest and under rotation, and the ability of the method to produce Campbell diagrams and stability maps at a fraction of the finite element cost. The framework extends directly to laminated composite sections by replacing the constitutive matrix, to arbitrary section families by replacing the camber and thickness functions, and to nonlinear prestress by iterating the equilibrium solve on the deformed geometry, none of which requires modification of the kinematic or discretization machinery.

## 7. Data Availability Statement

## 8. Conflict of Interest

## 9. Acknowledgements

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New added:

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